Help

Such is the relevance of the humble F1 key that it's the only function key of the 12 which has moved to the realm of being used as a verb. Now we have the Internet to solve all our issues. Well most of them. But back in the day F1 was the go-to key for quick solutions. Moving into the technology sphere we are constantly seeing innovation, but yet there are some areas that could do with a bit of help. Follow us as we try and find some solutions.

Battery Refresh Needed ASAP

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We had written about this in our last anniversary issue as well. So we thought it would be an appropriate time to get things in perspective about battery tech, as we have hardly seen any improvement since last year. As far as smartphones and tablets go, we are still using the same old Lithium-ion batteries.

One may argue that battery capacities are on the rise. But that's only logical isn't it? What else is going to drive those high-res, PPI heavy displays, the quad-core processors and those lovely graphics without you having to run to charge your mobile device every other hour? So, higher milli-ampere hour rating on the battery should never be taken to mean that the tech as such is improving. Innovation is happening on the system-on-chip front, with changes in micro-architectures which put less load on the battery life. But with our hunger for high resolution, high pixel-density screens, more graphics-heavy games, there is only so much you can innovate. At some point, optimisation will reach a limit and the only option left will be to improve on existing battery tech. Sure, we have found a way around this issue, partly. If you have noticed, the market these days is flooded with one accessory which everyone is claiming to be a life-saver – portable power banks. We agree that these Li-ion portable power chargers are impressive when you really need that extra bit of battery juice in emergency situations. Despite phone and tablets having greater battery capacities and optimised SoCs, there is barely a device which can go two days without needing a supplemental charge. Unless you are one of those rare people who have a smartphone but no Internet access, you can stretch the phone to two full days without a charge. In such cases, it helps if you have a portable charger around. But all said and done, that is still a roundabout way of addressing the issue at hand. With smartphones, it's a no-brainer, portable power banks can be easily carried around. When it comes to charging tablets, which have a higher battery capacity, the power banks get relatively heavier. Moreover, these are just solutions for mobile devices – a fraction of the possible devices using batteries. What about cameras? What about electric vehicles? Battery technology innovation needs to happen from the ground up. We came across three solutions that may just help.

3D microbatteries

Researchers at the University of Illinois have developed a microbattery array which employs 3D electrodes. The claim is that these batteries are much powerful than even the best supercapacitor and are capable of charging 1000 times faster